

SOFIA, BULGARIA · 12+ YEARS

Sherif Shalaby

Software Architect & Staff Engineer

I design and build production-grade distributed systems, enterprise platforms and AI applications, from architecture and technical strategy through implementation and delivery.

Java · Spring Boot · AI / RAG · Distributed Systems · Cloud · Kubernetes

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NOW	Software Architect at Avenga, architecture owner on a production AI platform
DOMAINS	Aviation, energy, government and defence, recruitment
OPEN TO	Staff, Principal, Architect, Engineering Lead

Case studies

AI Staffing Platform

ENTELLIGENCE · TECH LEAD & ARCHITECTURE OWNER

MAY 2026 - PRESENT
PROBLEM

An AI staffing product needed a retrieval-augmented assistant that answers from the organisation’s own corpus rather than from the model’s priors. It also had to be verifiable, because a plausible-sounding wrong answer is worse than no answer in a hiring context.

IMPACT

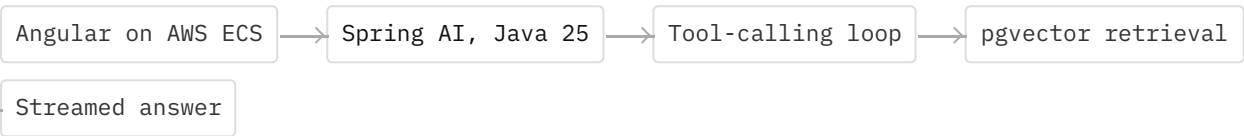
- A **one-button end-to-end harness** that boots a throwaway database and both applications for roughly **50 Playwright tests**. The project had no end-to-end verification before it.
- A **model-evaluation suite** scoring RAG answers for relevance and factuality, so answer quality is measured rather than assumed.
- A **~20-ticket technical-debt programme** delivered across both stacks.

ARCHITECTURE

Retrieval-augmented chat on **Java 25** and **Spring AI** over **PostgreSQL/pgvector**, streaming answers through a tool-calling loop, with an Angular front end on **AWS ECS** provisioned via **Terraform**. Third-party sync was moved out of the request transaction into a transactional outbox; session revocation and signed invitation and reset tokens were added.

MY ROLE

Own the architecture end to end. Set the engineering standards the team writes and reviews code against, including the AI-assisted review and planning workflow now shared across the repository. Own backlog planning, architecture overviews and code review.



Retrieval and generation sit on the request path. Third-party sync was moved off it into a transactional outbox.

Digital NOTAM Platform for EUROCONTROL

INDRA AVITECH · TEAM LEAD

AUG 2023 - DEC 2024
PROBLEM

EUROCONTROL needed a new Conventional and Digital NOTAM processing system: the notices that tell every pilot in European airspace what has changed. The system had to keep ingesting and distributing at full rate without interruption, and stay consistent with the air navigation systems around it.

100,000

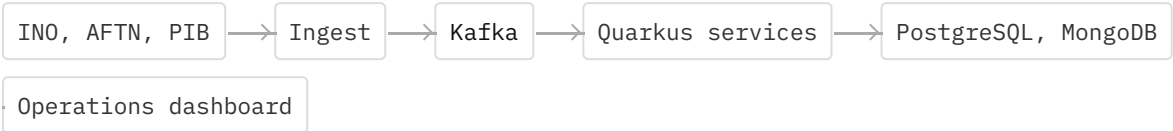
NOTAMS PER DAY, ENGINEERED FOR

24/7

AVAILABILITY TARGET

MY ROLE

Led the cross-functional team. Owned story analysis, sprint planning, code review and milestone delivery; set architectural standards and mentored engineers on quality and design.



Keycloak authenticates at the edge; the whole thing runs on OpenShift and Kubernetes.

ARCHITECTURE

Quarkus services with Kafka event streaming, PostgreSQL and MongoDB persistence, Keycloak authentication, deployed on OpenShift/Kubernetes. Integrated with the core aviation systems (INO, AFTN and PIB) for reliable exchange and operational consistency with external air navigation systems. ALSO DELIVERED

- A real-time operations dashboard for monitoring incoming NOTAM traffic and live system health.

ProMIS Modernisation

HITACHI ENERGY · SOFTWARE ARCHITECT

JUL 2025 - APR 2026
PROBLEM

A legacy enterprise platform on WildFly and JSF still ran the business, so it could not be switched off, rewritten in place, or paused for a migration. The work had to move it to a modern distributed architecture while the business kept operating on it. IMPACT

- The business stayed **fully operational throughout a phased migration**. The legacy application was containerised to run side by side with the new services on a shared database.
- Refactoring **cut the technical debt blocking new feature work**.

ARCHITECTURE

Migration from WildFly/JSF to Java 21 / Spring Boot 3.4 microservices, applying hexagonal architecture and an API-first, OpenAPI-generated contract layer. Multi-datasource transaction management, Liquibase schema delivery, Elasticsearch-backed search, and Keycloak/OIDC authentication behind an Nginx routing layer.

MY ROLE

Drove the migration and its architecture decisions, and raised the quality baseline with JUnit 5, Mockito, REST Assured and WireMock.



Both stacks served behind one routing layer against one database, which is what let the migration be phased instead of cut over.

TAKAFO Recruitment Platform

MUBADALA GROUP · ENGINEER, THEN SERVICE DELIVERY MANAGER

2021 - 2023, AND 2025
PROBLEM

A group-wide recruitment platform serving hiring across Mubadala Group companies, with a job-seeker portal, an internal UI and an admin portal for managing opportunities and matching. Later the engagement itself needed an owner accountable to the client.

IMPACT

- Every scheduled release delivered on time across the six-month delivery-ownership term, with no P1 incidents escaping to production.
- Kept the platform stable for hiring across Mubadala Group companies.

ARCHITECTURE

Backend and frontend of the platform (JobSeeker portal, Takafo UI and an admin portal for job opportunities and matching algorithms) with Elasticsearch queries powering recruitment analytics and statistical insight for HR stakeholders.

MY ROLE

Built the platform as an engineer, mentored junior and mid-level engineers, led code reviews and drove refactoring. Later ran a six-person delivery team as Service Delivery Manager, holding contractual accountability for SLA and uptime commitments, scope and change requests, and client steering and escalation, while continuing in the Software Architect role.



Elasticsearch queries powered the analytics and statistical insight HR stakeholders worked from.

Professional timeline

AVENGA, FORMERLY MUSALA SOFT

Jan 2021 - present, Sofia and Cairo

Software Architect

Architecture ownership across four client engagements (aviation, energy, recruitment and AI) while remaining hands-on in the code.

Software Architect 2025- Service Delivery Manager, TAKAFO 2025 (concurrent)

Associate Software Architect 2024 Senior Software Engineer 2021-23

TECHNICAL DETAIL

- Own the architecture of a production retrieval-augmented AI platform on Spring AI and pgvector, including its evaluation harness and engineering standards.
- Drove the ProMIS migration from WildFly/JSF to Java 21 and Spring Boot 3.4 microservices under hexagonal architecture and API-first contracts.
- Led the cross-functional team building EUROCONTROL’s Conventional and Digital NOTAM processing system on Quarkus, Kafka and OpenShift.
- Held contractual delivery accountability for TAKAFO as Service Delivery Manager: SLA and uptime commitments, scope and change requests, client steering and escalation.
- Musala Soft merged into Avenga in 2025; the architect track maps to Staff Software Engineer under Avenga’s job architecture.

Senior Software Engineer

Enterprise integration, workflow and ERP systems for government, defence and border-control clients, including an organisation-wide ESB.

Senior Software Engineer 2017-20 Associate Software Engineer 2014-17

TECHNICAL DETAIL

- Built **CONNECT**, an ESB integrating in-house ERP systems via Apache Camel and RabbitMQ, plus a runtime administration and request/response auditing console in Angular and Spring Boot.
- Designed and delivered an internal flight reservation platform for Saudi Border Guards: REST APIs for the mobile team, pessimistic and optimistic seat locking across a clustered deployment, Dijkstra-based shortest-path routing, and a “Flight Cure” admin tool to repair broken flights after outages.
- Implemented an organisation-wide workflow engine on Flowable (BPMN and CMMN), automating multi-level approval processes.
- Delivered ERP systems for military intelligence, employee case management and border case tracking using Java EE, SOA, JSF 2.1, JMS, PrimeFaces, Hibernate and Jasper Reports over Oracle 11g and MS SQL.
- Led technical design and database design, and managed deployments across WebLogic, Tomcat, TomEE and WildFly.

INDEPENDENT

Jul 2018 - Dec 2020

Listify, Product Owner and Lead Developer

Defined the product vision and led a five-person team building a cross-platform list, task and note application.

TECHNICAL DETAIL

- Built with Angular, Ionic, Cordova and Cloud Firestore.
- Ran the full Scrum cycle (backlog management, prioritisation and sprint reviews) while owning the technical design, API layer and UI.

CAIRO UNIVERSITY

2008 - 2013

BSc, Computer Engineering

Faculty of Engineering, Cairo University, Egypt.

CERTIFICATIONS & LANGUAGES

- SystemsExpert, algoexpert.io (2023)
- Intro to DevOps, Udacity (2022)
- Building a Search Engine, Udacity (2021)
- Divide and Conquer, Sorting and Searching, and Randomized Algorithms, Coursera (2021)
- Angular, The Complete Guide, Udemy (2020)
- Languages: Arabic (native), English (professional working), French (limited working)

Domains

Architecture

Distributed Systems

Microservices

Event-Driven Architecture

Backend

JavaSpring BootSpring AI

QuarkusRESTKafka

AI

RAGLLM Applications

Tool CallingVector Search

AI Evaluation

DDDHexagonal ArchitectureSystem Integration

Production AI Architecture

FrontendAngularTypeScriptRxJS

DataPostgreSQLpgvectorMongoDBElasticsearch

Cloud & PlatformAWSKubernetesOpenShiftDockerTerraform

SecurityKeycloakOAuth2OIDCSAMLSAML

TestingJUnit 5MockitoREST AssuredWireMockPlaywrightTDD

Things I build outside work

NPM · APACHE-2.0

keeldocs

Test coverage for your documentation. It ties each documented claim to the code it describes and reports which claims stopped being true, with the evidence for each.

PRODUCT · 2018-2020

Listify

A cross-platform list, task and note application. I defined the product vision and led a five-person team as Product Owner.

MIT · LIVE DEMO

VibeMesh-AI

Describe a part in plain language and get a parametric, print-ready 3D model, rendered entirely in the browser.

THIS SITE

sherif-cv

This portfolio, and the interactive flight-strip version behind it: an air-traffic-control strip bay rendered procedurally in Three.js.

Let’s build something difficult.

AVAILABLE FOR

Staff Engineer, Principal Engineer, Software Architect, Engineering Lead